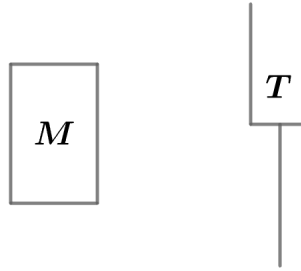


2009 F=ma Exam: Problem 17

Kevin S. Huang



- A) The dimension of g is L/T^2 . Since you don't have any tools to measure length in SI units, you can't measure g in SI units.
- B) The dimension of c is L/T . Since you don't have any tools to measure length in SI units, you can't measure c in SI units.
- C) The dimension of ρ is M/L^3 . Since you don't have any tools to measure length in SI units, you can't measure ρ in SI units.
- D) The dimension of k is M/T^2 . Since your mass and tuning fork can measure mass and time in SI units, you can measure k in SI units.
- E) The dimension of P is M/LT^2 . Since you don't have any tools to measure length in SI units, you can't measure P in SI units.

Thus, the answer is D.